Department of Education

STEM Education
National Schools of Excellence
Physics Syllabus
Grade 12
Term 1

DOMAIN	Phylum
SCIENCE	Physics
Technology	
Engineering	
Mathematics	

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SECRETARY'S MESSAGE

This Physics Syllabus is to be used exclusively by teachers of Physics to teach Grade 12 Students enrolled for the STEM Education through the Schools of Excellence throughout Papua New Guinea. The Grade 11 STEM Physics Syllabus ensured that Grade 11 students enrolling for STEM Education were grounded in the basic concepts espoused in Physics before undertaking the advanced concepts introduced in this syllabus.

This syllabus conforms to the visions of the National Education Plan and the School of Excellence Policy. Furthermore, it is undergirded by a policy and legislative matrix that were issued by the National Government over the last decades, such as the Vision 2050 and Development Strategic Plan 2030. The Preamble of the National Constitution, especially the First Directive of the *National Goals and Directive Principles* makes a clarion call for the Integral Human Development, which the Department took heed of to ensure, through this new syllabus, that *all forms of beneficial creativity, including sciences and cultures, [are] to be actively encouraged*. A principal object of the *National Education Act 1983 (amended 1995)* reiterates this virtue of Integral Human Development (Section 4).

In accordance with the widespread demand for a home-grown curriculum to replace the existing one with the view to position the country as a bastion of innovation in the new century the Secretary of Education has embarked on introducing the STEM Curriculum to the Schools of Excellence. It is hoped that students coming through the STEM Education will not only gain productive employment in the knowledge-based economy but become creators, innovators and inventors of knowledge and ultimately commercialize their creativity through development of their intellectual property rights.

In addition, there are four core elements espoused in this syllabus, insofar as student development is concerned: Students are expected to gain knowledge and Skills and apply them to solve every day problems whilst appreciating Physics as a fundamental field of science.

This STEM Physics Syllabus for Grade 12 constitutes six out of the ten (10) Thematic Area (Thema) and is intended for students enrolling for STEM Education. In this booklet only three Thema are presented. By studying these Thema students are expected to gain the foundational knowledge needed in Physics.

The success of this new Curriculum is highly dependent on the Teachers. They play a pivotal role by being innovative and keeping abreast of new information on scientific research and innovations. The challenge for teachers of Physics is to engage students to learn realistic context for increased and better Understanding and appreciate the relevance of Physics to human lives, including good health and environment.

In order to ensure that teachers discharge their responsibilities adequately this syllabus is accompanied by a Teacher's Guide and Teacher's Resource Book which provides sufficient teaching and assessment materials to help Teachers.

I commend and approve this syllabus as the official curriculum for Physics to be used for the Grade 12 STEM Education in the Schools of Excellence throughout Papua New Guinea.

DR UKE KOMBRA, PHD
Secretary for Education

INTRODUCTION

Physics is the natural science that studies matter, its fundamental constituents, its motion and behaviour through space and time, and the related entities of energy and force. It is used to study how the universe behaves.

Since Physics is an experimental science, laboratory work becomes an essential part of the syllabus. Practical work can be used to enable students to investigate the properties and reactions of substances and to provide opportunities for students to learn and test the principles and concepts of Physics.

The Grade 12 Physics Syllabus entails six (6) of ten (1) Thematic Areas (Thema) and these are listed below. Each Thema has a single Module and three topics. Over time different Modules will be introduced.

Since Physics is a highly specialized subject students undertaking Physics are expected to possess high levels of cognitive and numeracy competency, and a firm grasp of the language skills.

The modules in this Syllabus are sequenced to guide teachers of Physics through the process of imparting basic Physics knowledge and skills to students.

THEMA	MODULE	TOPICS
BIOTECHNOLOGY	12.1: Biophysics	1. Introduction to Biophysics 2. Genomics – DNA Sequencing 3. Proteomics
	12.2: Bioelectronics	1. Applications of Bioelectronics
HEALTH AND MEDICINE	12.1: Nuclear Medicine	1. Introduction to Nuclear Medicine 2. Nuclear Medicine – Diagnosis 3. Nuclear Medicine Therapy
ENERGY	12.1: Nuclear Energy	1. Introduction to Nuclear Energy 2. Nuclear Reactions – Fission and Fusion 3. Nuclear Fission Reactors
BATTERY TECHNOLOGY	12.1: Inductive Charging	1. Electromagnetism 2. Inductive Charging 3. Hybrid Electric Vehicles
ARTIFICIAL INTELLIGENCE	12.1: Data and Artificial Intelligence	1. Introduction to Artificial Intelligence 2. Machine Learning 3. Neural Networks 4. Data Mining
ROBOTICS	12.1: Electronics	1. Robotics and Electronics 2. Control Systems 3. Circuit Boards

Since Physics is a highly specialized subject students undertaking Physics are expected to possess high levels of cognitive and numeracy competency, and a firm grasp of the language Skills.

The modules in this Syllabus are sequenced to guide teachers of Physics through the process of imparting basic Physics knowledge and Skills to students.

RATIONALE

There is a confluence of events that necessitated the need to introduce STEM Curriculum to the Schools of Excellence:

1. The School of Excellence Policy that was launched by Prime Minister Honourable James Marape, MP, in June 2020, calls for the establishment of the Schools of Excellence and the introduction of STEM Curriculum;

2. The Government has decided to replace the Observation Based Education to Standards Based Education in 2019 and tasked the Department of Education to introduce the STEM Curriculum as part of its reform agenda;

3. There is public outcry, both from public and private sectors, over the quality of graduates coming through the National Education System; and

4. The changing landscape in the technology domain globally and the pressing need to produce the next generation of workforce that is agile and can adopt to the changing circumstances as well as become enterprising individuals through their creation of intellectual properties.

The combination of these factors has led the Department of Education to adopt the STEM Curriculum and to introduce this to the Schools of Excellence as a necessary disruptive intervention in an effort to develop the next generation of national intellectual capital and to drive the growth and prosperity of the country through creation of intellectual properties.

In the STEM Grade 11 Physics Syllabus, students were introduced to the basic concepts and knowledge of Physics in order for them to develop an appreciation of the discipline in a broader context. In this syllabus students are introduced to advanced concepts in Physics as they are applied to six Thema.

The introduction of the STEM Curriculum will cause consequential changes to the Upper Classes of STEM Education and the Lower Secondary Syllabi. These syllabi will be introduced in the course of time.

NEW NOMENCLATURE

As part of the overall Curriculum Development, new nomenclature was introduced. These are explained below:

Term	**Explanation**
Domain	Each STEM area is defined as *domain* (e.g., Science Domain, Technology Domain, etc.)
Phylum	What would normally be regarded as "subject" (e.g., Physics, Physics, Physics, etc.) is now regarded as *phylum*; the plural is *phyla*.
Thema	Thematic areas of studies
Module	A specific concept within the Thema which is being studied
Topic	A specific aspect of the Module that is being studied

STEM Domains

At this juncture of our national development there is no other task before us more important than deploying a robust STEM Curriculum that develops the next generation of highly educated students to enter the work force. By building on the best of current practices and standards the STEM Curriculum aims to take us, beyond the constraints of the existing structures of schooling, towards a shared vision of excellence nationally and marketing the nation globally in terms of Science, Technology, Engineering and Mathematics. In this STEM Curriculum, there are ten (10) Thema (Thematic Areas) introduced.

Science
It is the study of the natural world, including the laws of the nature associated with physics, Physics and Physics and the treatment or application of facts, principles, concepts or conventions associated with these disciplines. Science is both a body of knowledge that has been accumulated over time and a process-scientific inquiry-that generates new knowledge. Knowing from science informs the engineering design process. In this STEM Curriculum the Science Domain has three Phyla – Physics, Physics and Physics.

Technology
Comprises the entire system of people and organizations, knowledge, processes, and devices that go into creating and operating technological artifacts, as well as the artifacts themselves. Throughout history humans have created technology to satisfy their wants and needs. Much of modern technology is the product of science and engineering and technological tools used in both fields. There are two Phyla – Information Technology and Computer Sciences – introduced in this STEM Curriculum for the Technology Domain.

Engineering
Concerns both the knowledge about the design and creation of products and a process for solving problems. The design process must result in a final product that must operate or function under various constraints. One constraint in engineering design is the laws of nature. Engineers must learn to harness the natural forces at work around them and through their design process develop products that must operate in harmony with nature. Other constraints include things such as time, money, available materials, ergonomics, environmental regulations and manufacturability. Engineering utilizes concepts in science and mathematics as well as technological tools. There are three (3) Phyla introduced in the Engineering Domain – Civil, Mechanical and Electrical.

Mathematics
It is the study of patterns and relationships among quantities, numbers, and shapes. Specific branches of mathematics include arithmetic, geometry, algebra, trigonometry and calculus. Mathematics is used in science and in engineering. In this STEM Curriculum, Mathematics is considered as a Service Domain; specific topics tailored towards specific Thema are also considered relevant to undergird the relevance and applicability of Mathematics Domain to other fields of knowledge.

STEM GRADE 12 PHYSICS CONTENT OVERVIEW – Term 1

Table 1.1 Sequencing & Benchmarking STEM Physics Content Standards

Setting STEM Physics Content Standards for Progressive Learning & Assessment			
Thema BIOTECHNOLOGY	**Module Content Standards**		
Module 12.1: Biophysics	**12.1:** Conceptualize the significant contribution of physics in biophysics and identify its practical concepts that can be applied to solve problems in the real world.		
Topics	**Performance Indicators**	**Module Content Benchmark**	**Assessment Tasks**
1: Introduction to Biophysics	**12.1.1a:** Clearly define what biophysics and explain how it is as an integrated field of science. **12.1.1b:** Understand and differentiate the different applications in biophysics. **12.1.1c:** Differentiate between Genomics and Proteomics. **12.1.1d:** Describe the significant roles of Genomics and Proteomics in biophysics.		Research Project + Written Test
2: Genomics – DNA Sequencing	**12.1.2a:** Clearly define and describe Genomics and relate its concepts to physics. **12.1.2b:** Evidently distinguish between what a gene and genome is. **12.1.2c:** Highlight and articulate what DNA sequencing and Sanger Sequencing is. **12.1.2d:** List and describe the different ingredients used in Sanger Sequencing. **12.1.2e:** Explain and describe clearly the basic methodology and processes		

	involved in performing Sanger Sequencing. **12.1.2f:** Understand and explain what Bioinformatics is.		
3: Proteomics	**12.1.3a:** Clearly differentiate between a protein and proteome. **12.1.3b:** Clearly outline and explain the relationship between Proteomics and Genomics. **12.1.3c:** Describe the process involved in Edman Degradation. **12.1.3d:** Compare the process of Edman Degradation and Sanger Sequencing.		
Module 12.2: Bioelectronics	**12.2:** Determine and highlight the role of bioelectronics that assist in creating devices for improving human deficiencies.		
Topics	**Performance Indicators**	**Module Content Benchmark**	**Assessment Tasks**
1: Applications of Bioelectronics	**12.2.1a:** define bioelectronics and its significance to biotechnology. **12.2.1b:** Clearly explain how the heart works and describe how the pacemaker is applied for patients with cardiac problems. **12.2.1c:** Clearly explain how the ear works and describe how the hearing aid assists patients with hearing problems.	Students' ability to master the knowledge and Skills of: • Recognizing and identifying bioelectronics that are currently used; and • Explaining the physics related to using bioelectronics in current real-life applications.	Research Project + Written Test

Thema Health and Medicine	**Module Content Standards**
Module 12.1 Nuclear Medicine	**12.1:** Learn, recognize and comprehend the physics involved in using radiopharmaceuticals and the different nuclear emission tomographs used for medical diagnosis and therapy.

Topics	Performance Indicators	Module Content Benchmark	Assessment Tasks
1: Introduction to Nuclear Medicine	**12.1.1a:** Define nuclear medicine as a combination of radiology, medicine and molecular biology. **12.1.1b:** Explain the general knowledge of physics utilized in nuclear medicine. **12.1.1c:** Explain what radiopharmaceuticals are, their production, and how it is administered to patients. **12.1.1d:** Assess and highlight the risks for involved in using nuclear medicine.	Students' ability to master the knowledge and Skills of; • Explaining the basic production and administering of radiopharmaceuticals; and • Distinguishing the technologies and processes involved in nuclear diagnosis and therapy.	Research Project + Written Test
2: Nuclear Medicine - Diagnosis	**12.1.2a:** Clearly define nuclear medicine diagnosis and emission tomography. **12.1.2b:** Highlight and generally explain the concept of radiation used in PET and SPECT scans. **12.1.2c:** Clearly differentiate the process and methods used in PET and SPECT scans. **12.1.2d:** Compare and contrast PET and SPECT scans and their effectiveness in medical diagnosis.		
3: Nuclear Medicine - Therapy	**12.1.3a:** Clearly define nuclear medicine therapy and explain in contrast to nuclear diagnosis. **12.1.3b:** Describe the physical and biochemical characteristics for choosing and administering a radionuclide. **12.1.3c:** Describe how radioactive Iodine therapy, treatment of bone metastases and radio		

	immunotherapy is conducted.		

Table 1.2 Mapping Content Progressions by Modules

Thema/Module	Module Content Standards	12.1	12.2	12.1
BIOTECHNOLOGY / Module 12.1: Biophysics	**12.1:** Conceptualize the significant contribution of physics in biophysics and identify its practical concepts that can be applied to solve problems in the real world.	✓		
BIOTECHNOLOGY / Module 12.2: Bioelectronics	**12.2:** Determine and highlight the role of bioelectronics that assist in creating devices for improving human deficiencies.		✓	
HEALTH AND MEDICINE / Module 12.1: Nuclear Medicine	**12.1:** Learn, recognize and comprehend the physics involved in using radiopharmaceuticals and the different nuclear emission tomographs used for medical diagnosis and therapy.			✓

THEMA BIOTECHNOLOGY

Introduction

Biotechnology is a branch of biology and chemistry that employs mathematics and physics to develop tools and engineering insights for modern biology and biomedical research. It entails the study and application of living cells and cellular materials in the development of pharmacological, diagnostic, agricultural, environmental, and other products that promote human welfare and society. It's also utilized to examine and manipulate genetic information in animals in order to imitate and study human disease. Biotechnology is a science that is both fundamental and applied. Agriculture, health care and medical sciences, biomedical engineering, cell biology, immunology, seed technology, virology, plant physiology, forensics, industrial processing, and environmental management all benefit from the technologies it develops.

There are a number of career paths linked to the field of Biotechnology, such as: working in research labs, graduate training in biology and chemistry, and professional careers in the health sciences. If you are someone who enjoys STEM (particularly biology and chemistry), below are some best biotechnology careers that may be of interest to you (research on them for more information):

- Biomedical Engineer
- Biochemistry
- Medical Scientist
- Clinical Technician
- Microbiologist
- Process Development Scientist
- Biomanufacturing Specialist
- Business Development Manager
- Product Strategist
- Biopharma Sales Representative
- Medical Scientist
- Biotechnological Technician
- Epidemiologist
- Microbiologist
- Medical and Clinical Lab Technologist
- Biomanufacturing Specialist
- Bioproduction Specialist

MODULE 12.1: BIOPHYSICS

4-5 Weeks

Biophysics is a vibrant scientific field of study that comprises a variety of disciplines including mathematics, chemistry, engineering, material science and other life sciences. The primary aim of this field is to utilize and apply the principles and laws of physics to understand how biological systems work.

In this module, the basic concepts of biophysics will be covered that describes how this field acts as a bridge between the science domains (i.e., Physics, Biology and Chemistry). This will establish a clear Understanding to then discuss its influence as a budding field of science, and its significant contributions to proteomics (protein science) and genomics (genetics) which have broader applications in biotechnology.

Linking Modules

- Grade 11 Term 2, STEM Engineering (Thema Biotechnology)
- Grade 12 Term 1, STEM Biology (Thema Biotechnology)
- Grade 12 Term 1, STEM Chemistry (Thema Biotechnology)

Module Content Standard

12.1: Conceptualize the significant contribution of physics in biophysics and identify its practical concepts that can be applied to solve problems in the real world.

Module Content Benchmark

Students' ability to master the knowledge and Skills of:

- Recognize and understand biophysics as an integrated field of science.
- Fundamental concept of Genomics (DNA Sequencing) and Proteomics (Protein Sequencing).
- Sanger Sequencing.
- Edman Degradation Process.

Topic 1: Introduction to Biophysics

Performance Indicators

Students are expected to:

12.1.1a: Clearly define what biophysics and explain how it is as an integrated field of science.
12.1.1b: Understand and differentiate the different applications in biophysics.
12.1.1c: Differentiate between Genomics and Proteomics.
12.1.1d: Describe the significant roles of Genomics and Proteomics in biophysics.

Content Expansion

Body of Knowledge for their Understanding

- Introduction
    - Definition
    - Biophysics: The Bridging Science
- Applications of Biophysics
    - DNA Structure
    - Computer Modelling
    - Molecules in Motion
    - Neuroscience
    - Medical Applications
        - Biomechanics
        - Nuclear Medicine
    - Genomics and Proteomics
        - Differences between Genomics and Proteomics

Set of Skills for Proficiency Levels

- Develop a basic Understanding of the concepts and methods involved with biophysics
- Analytical problem solving
- Research Skills
- Written and oral communication Skills
- Observational Skills
- Teamwork and interpersonal Skills

Traits of Attitudes and Values to govern their lives

- Appreciate the role of biophysics and its applications in the modern technological age.
- Develop a logical and independent mind.
- Meticulous attention to detail and accuracy.

Topic 2: Genomics – DNA Sequencing

Performance Indicators

Students are expected to:

12.1.2a: Clearly define and describe Genomics and relate its concepts to physics.
12.1.2b: Evidently distinguish between what a gene and genome is.
12.1.2c: Highlight and articulate what DNA sequencing and Sanger Sequencing is.
12.1.2d: List and describe the different ingredients used in Sanger Sequencing.
12.1.2e: Explain and describe clearly the basic methodology and processes involved in performing Sanger Sequencing.
12.1.2f: Understand and explain what Bioinformatics is.

Content Expansion

Body of Knowledge for their Understanding

- Introduction
- DNA and DNA Sequencing
    - Brief History of DNA Sequencing
    - Sanger Sequencing
        - Ingredients
            - Template Strand
            - DNA Polymerase
            - DNA Primer
            - Deoxy Nucleotides
            - Dideoxy Nucleotides
- Basic Methodology
    - Obtaining the template strand
    - Addition of a Primer and dispersing into reaction tubes
    - Addition of DNA Polymerase, Deoxy and Dideoxy Nucleotides
    - DNA Replication and Termination
    - Denaturing the DNA fragments into single strand DNA
    - Gel Electrophoresis
- Bioinformatics

Set of Skills for their Proficiency Levels

- Apply the methods of DNA extraction and sequencing in laboratory experiments.
- Research Skills
- Written and oral communication Skills
- Observational Skills
- Teamwork and interpersonal Skills

Traits of Attitudes and Values to govern their lives to govern their lives

- Appreciate the contribution of Genomics in real life applications.
- Develop a logical and independent mind.
- Meticulous attention to detail and accuracy.

Topic 3: Proteomics

Performance Indicators
Students are expected to:
12.1.3a: Clearly differentiate between a protein and proteome.
12.1.3b: Clearly outline and explain the relationship between Proteomics and Genomics.
12.1.3c: Describe the process involved in Edman Degradation.
12.1.3d: Compare the process of Edman Degradation and Sanger Sequencing.

Content Expansion
Body of Knowledge for their Understanding
- Introduction
- Gene expression and Protein Sequencing
    - Edman Degradation Process
    - Relation of the processes involved with Edman degradation and Sanger Sequencing.

Set of Skills for their Proficiency Levels
- Discover and apply the methods of DNA extraction and sequencing using BLAST.
- Research Skills
- Written and oral communication Skills
- Observational Skills
- Teamwork and interpersonal Skills

Traits of Attitudes and Values to govern their lives to govern their lives
- Appreciate the contribution of Proteomics in real life applications.
- Develop a logical and independent mind.
- Meticulous attention to detail and accuracy.

MODULE 12.2: BIOELECTRONICS

4-5 Weeks

Bioelectronics utilizes electronic devices to monitor basic biological functions and is particularly used to assist biological systems in humans that are diagnosed to be defective or underperforming. These electronic devices help to modulate, initiate and trigger the response that will otherwise naturally be triggered by biological processes.

In this module, we will discuss some key applications of bioelectronics that are commonly used that will illuminate the mindset of students to investigate into how the concepts of physics can have biological applications.

Module Content Standard

12.2: Determine and highlight the role of bioelectronics that assist in creating devices for improving human deficiencies.

Module Benchmark

Students' ability to master the knowledge and Skills of:

- Recognizing and identifying bioelectronics that are currently used; and
- Explaining the physics related to using bioelectronics in current real-life applications.

Topic 1: Applications of Bioelectronics

Performance Indicators

Students are expected to:

12.2.1a: define bioelectronics and its significance to biotechnology.

12.2.1b: Clearly explain how the heart works and describe how the pacemaker is applied for patients with cardiac problems.

12.2.1c: Clearly explain how the ear works and describe how the hearing aid assists patients with hearing problems.

Content Expansion

Body of Knowledge for their Understanding

- Introduction
  - Definition
- Applications of Bioelectronics
  - Cardiac Pacemaker
    - Basic Anatomy and Electrophysiology of the Heart
    - Mechanism of the Cardiac Pacemaker
  - Hearing Aid
    - How the Ear works
      - Outer Ear
      - Middle Ear
      - Inner Ear
    - Mechanism of the Hearing Aid

Set of Skills for their Proficiency Levels

- Discover the applicable roles of bioelectronics that aid in human deficiencies.
- Research Skills
- Written and oral communication Skills
- Observational Skills
- Teamwork and interpersonal Skills

Traits of Attitudes and Values to govern their lives to govern their lives

- Appreciate and value the applications of bioelectronics in real world applications.
- Develop a logical and independent mind.
- Meticulous attention to detail and accuracy.

THEMA HEALTH AND MEDICINE

Introduction

The study of the human mind and body, how they function, and how they interact - not only with one another but also with their surroundings - has always been critical to human well-being. With the advancement of modern medicine, research on new therapies and preventative medicine has exploded, and a network of disciplines, including genetics, psychology, and nutrition, is working to improve human health.

MODULE 12.1: NUCLEAR MEDICINE

4-5 Weeks

Nuclear medicine is the science and clinical practice that involves the usage of radioactive compounds called radiopharmaceuticals (or more commonly called radiotracers) that are administered into the body. The radiation emitted is captured by certain medical instruments and assist nuclear medical practitioners to identify different types of diseases that can range from cancer, heart diseases and neurological disorders.

In this module, we will discuss the Physics associated with Nuclear Medicine. We will begin by Understanding the nucleus of atoms at the sub-atomic level and how it relates to certain elements and chemical compounds becoming radioactive. This will serve as the basis to Understanding radiotracers used in nuclear medicine and will assist students to understand how these instruments administer and use radiotracers to help identify and prevent the spread of the medical conditions mentioned above.

The main focus of Topic 1 is on Radionuclides used in Nuclear Medicine. Topic 2 focusses on Imaging modalities (SPECT and PET Scans). Topic 3 focusses on Therapeutic applications emphasizing on serious treatments such as cancer treatment and bone metastasis.

Linking Modules

- Grade 11 Term 3, STEM Physics (Thema: Health and Medicine)

Module Content Standard

12.1: Learn, recognize and comprehend the physics involved in using radiopharmaceuticals and the different nuclear emission tomographs used for medical diagnosis and therapy.

Module Content Benchmark

Students' ability to master the knowledge and Skills of;

- Explaining the basic production and administering of radiopharmaceuticals; and
- Distinguishing the technologies and processes involved in nuclear diagnosis and therapy.

Topic 1: Introduction to Nuclear Medicine

Performance Indicators

Students are expected to:

12.1.1a: Define nuclear medicine as a combination of radiology, medicine and molecular biology.

12.1.1b: Explain the general knowledge of physics utilized in nuclear medicine.

12.1.1c: Explain what radiopharmaceuticals are, their production, and how it is administered to patients.

12.1.1d: Assess and highlight the risks for involved in using nuclear medicine.

Content Expansion

Body of Knowledge for their Understanding

- Understanding Nuclear Medicine
- How Nuclear Medicine Works
- Radiopharmaceuticals
  - Technium-99m
  - Production
  - Administration and Mechanism of Action
- Advantages of Nuclear Medicine
- Disadvantages of Nuclear Medicine

Set of Skills for Proficiency Levels

- Discover and identify the solutions provided by using radiation from radioactive substances and technologies in nuclear medicine.
- Research Skills
- Written and oral communication Skills
- Observational Skills
- Teamwork and interpersonal Skills

Traits of Attitudes and Values to govern their lives

- Appreciate the fundamentals of physics used in nuclear medicine and set them as the building blocks for advanced topics under nuclear medicine.
- Develop a logical and independent mind.
- Meticulous attention to detail and accuracy.

Topic 2: Nuclear Medicine - Diagnosis

Performance Indicators

Students are expected to:

12.1.2a: Clearly define nuclear medicine diagnosis and emission tomography.

12.1.2b: Highlight and generally explain the concept of radiation used in PET and SPECT scans.

12.1.2c: Clearly differentiate the process and methods used in PET and SPECT scans.

12.1.2d: Compare and contrast PET and SPECT scans and their effectiveness in medical diagnosis.

Content Expansion

Body of Knowledge for their Understanding

- Introduction
- Emission Tomography
- Single Photon Emission Computerized Tomography
    - Differences to other Imaging Techniques
    - Purpose and Uses of SPECT scans
        - Neuroimaging
        - Cardiac imaging
        - Skeletal Imaging
- Positron Emission Tomography
    - Purpose and uses of PET scans
- Comparison of PET and SPECT scans

Set of Skills for Proficiency Levels

- Research Skills
- Written and oral communication Skills
- Observational Skills
- Teamwork and interpersonal Skills

Traits of Attitudes and Values to govern their lives

- Appreciate the fundamentals of physics used in nuclear medicine and set them as the building blocks for advanced topics under nuclear medicine.
- Develop a logical and independent mind.
- Meticulous attention to detail and accuracy.

Topic 3: Nuclear Medicine Therapy

Performance Indicators
Students are expected to:
12.1.3a: Clearly define nuclear medicine therapy and explain in contrast to nuclear diagnosis.
12.1.3b: Describe the physical and biochemical characteristics for choosing and administering a radionuclide.
12.1.3c: Describe how radioactive Iodine therapy, treatment of bone metastases and radio immunotherapy is conducted.

Content Expansion
Body of Knowledge for their Understanding

- Introduction
- Clinical requirements and choice of Radionuclides
  - Physical Characteristics
    - Physical Half-life
    - Types of Emission
    - Energy in Radiation
    - Daughter Products
  - Biochemical Characteristics
    - Tissue Targeting and Retention of Radioactivity
    - In Vivo Stability
    - Toxicity
- Selected Therapeutic Radionuclides and their Clinical Applications
  - Radioactive Iodine Therapy
  - Treatment of Bone Metastases
  - Radio Immunotherapy
    - ^{90}Y-ibritumomab tiuxetan
    - ^{131}I-tositumomab
    - Epratuzumab

Set of Skills for Proficiency Levels

- Research Skills
- Written and oral communication Skills
- Observational Skills
- Teamwork and interpersonal Skills

Traits of Attitudes and Values to govern their lives

- Appreciate the knowledge of physics behind nuclear medicine imaging devices.
- Develop a logical and independent mind.
- Meticulous attention to detail and accuracy.

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